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# Timing Belt Tension Adjustment

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## Pre-Operation Information

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### 1.1 Purpose

This SOP outlines how to adjust the X/Y-axis timing belts of the FibreSeeker 3 3D printer. It resolves issues including belt slippage caused by insufficient tension, skipped-tooth noise, homing collision, and print position deviation to prevent accelerated wear of motors and bearings arising from excessive belt tension.

### 1.2 Scope

Applicable to the installation, tension inspection, and adjustment of the X- and Y-axis timing belts of the FibreSeeker 3.

### 1.3 Notes

It is necessary to inspect, adjust, and test the belt in accordance with the procedure; record the printer status before and after tuning; report any abnormalities that cannot be resolved promptly.

## 1.4 Required Tools

Tensiometer, hex key, feeler gauge, screwdriver



## Troubleshooting Steps

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### Step 1: Power Off

Disconnect the printer's power cord. Do not perform any adjustment or tightening of fasteners while the printer is powered on.



### Step 2: Set Parameters

Configure the tensiometer with the following parameters:

- M=000.9 g/m
- W=006.0 mm/R
- S=0360 mm



### Step 3: Check Belt Tension

Move the print head to the far left. Apply vertical pressure at the midpoint of the belt span with a tensiometer, and keep the tension value within the range of 12~20.



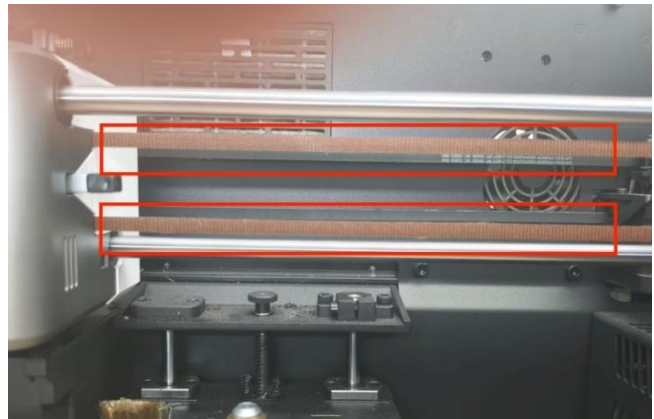
If no tensiometer is available, press the belt manually; obvious slackness indicates insufficient belt tension.



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### Step 4: Confirm the Adjustment Position

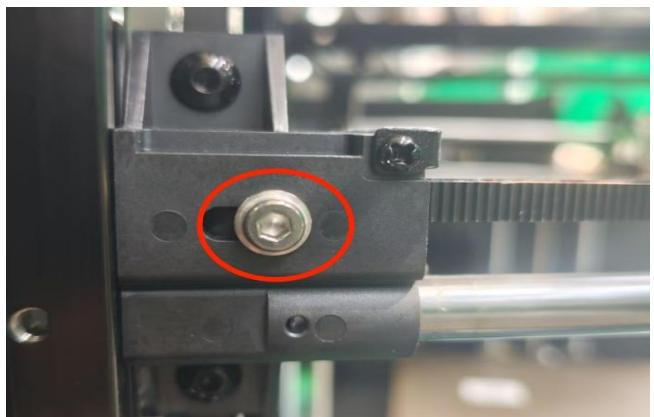
There are two belts vertically: the upper belt's adjustment point is on the left, while the lower belt's adjustment point is on the right side.



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### Step 5: Locate the Fixing Screws

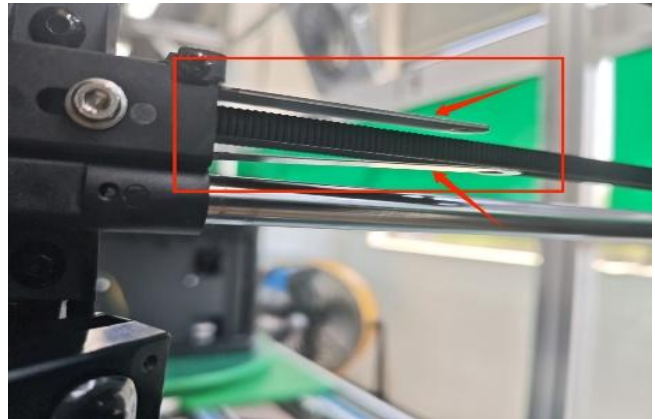
Locate the fixing screws at the outer end of the belt and confirm the target screws to be adjusted.



## Step 6: Adjust

Insert feeler gauges into the top and bottom gaps of the idler pulley:

- Left: 1.00 mm feeler gauge top, 0.4 mm bottom
- Right: 1.00 mm feeler gauge bottom, 0.4 mm top



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## Step 7: Tension Adjustment

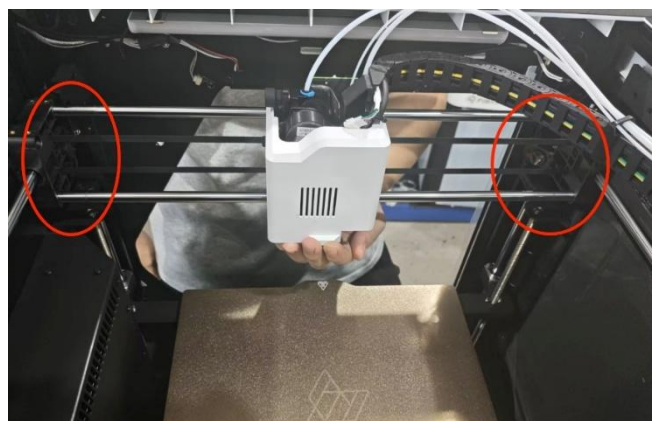
Loosen the fixing screws with a hex key; hold the tensioning block fixed with a screwdriver. Push the block inward to increase belt tension, or pry it outward to reduce tension.



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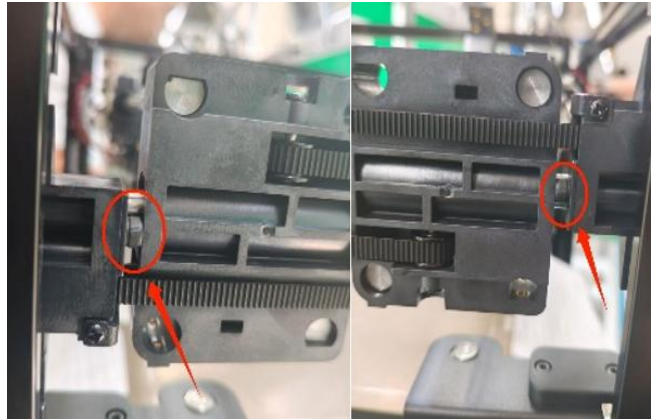
## Step 8: Level Assemblies

Move the print head to the middle of the printer. Shift the assemblies to the far left and far right ends of the Y-axis. Manually move the print head to check whether



the left and right assemblies are at the same height.

If it is difficult to judge whether the assemblies are level, inspect the balance screws on both sides.



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### Step 9: Recheck Alignment

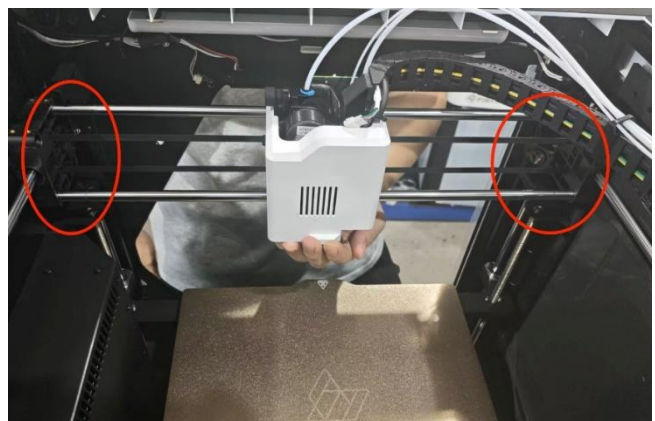
If the left and right assemblies are clearly misaligned, carry out secondary adjustment: tighten the screws on the uneven side accordingly as shown in the picture.



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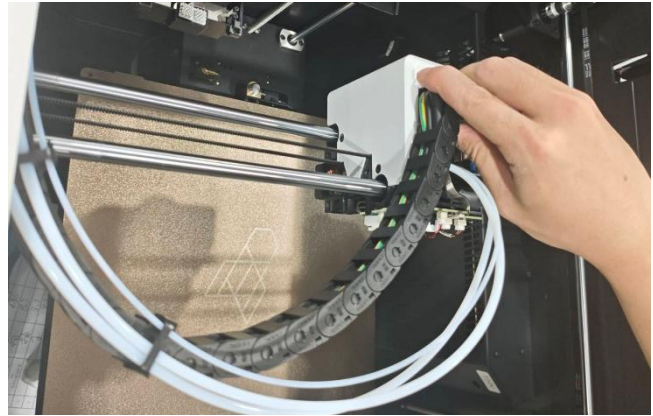
### Step 10: Confirm Alignment

Manually move the print head to recheck whether the left and right assemblies are level and confirm the alignment result.



## Step 11: Homing Calibration Test

1. Hold the print head and move it forward, backward, leftward and rightward for inspection.
2. Enter 0 for both X and Y coordinates to trigger the printer's automatic X/Y-axis homing procedure. Inspect the following conditions:
  - No tooth skipping, slipping or impact noise occurs when the print head triggers the limit switches.
  - No persistent clicking or slipping noise during printer operation.



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